

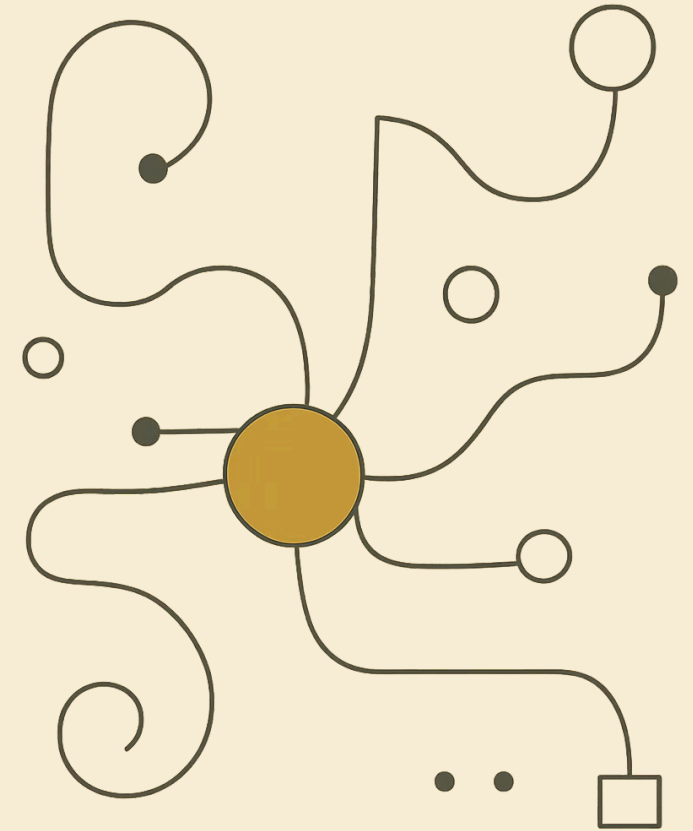
Customer Churn Analysis & Revenue Impact Modelling

An end-to-end data analytics portfolio project analysing telecom customer churn using **Python**, **SQL**, and **Tableau** — turning raw customer data into business-ready retention strategies.

PORTFOLIO PROJECT

TELECOM ANALYTICS

PYTHON · SQL · TABLEAU





Why Customer Churn Matters

Customer churn is one of the most costly challenges in the telecom industry. This project analyses **3,676 customers** across **44 features** to uncover patterns, quantify financial impact, and identify at-risk segments.

50.84%

Overall Churn Rate

More than half of all customers
churned during the analysis period

\$3.68M

Revenue Lost

Total revenue lost due to churned
customers

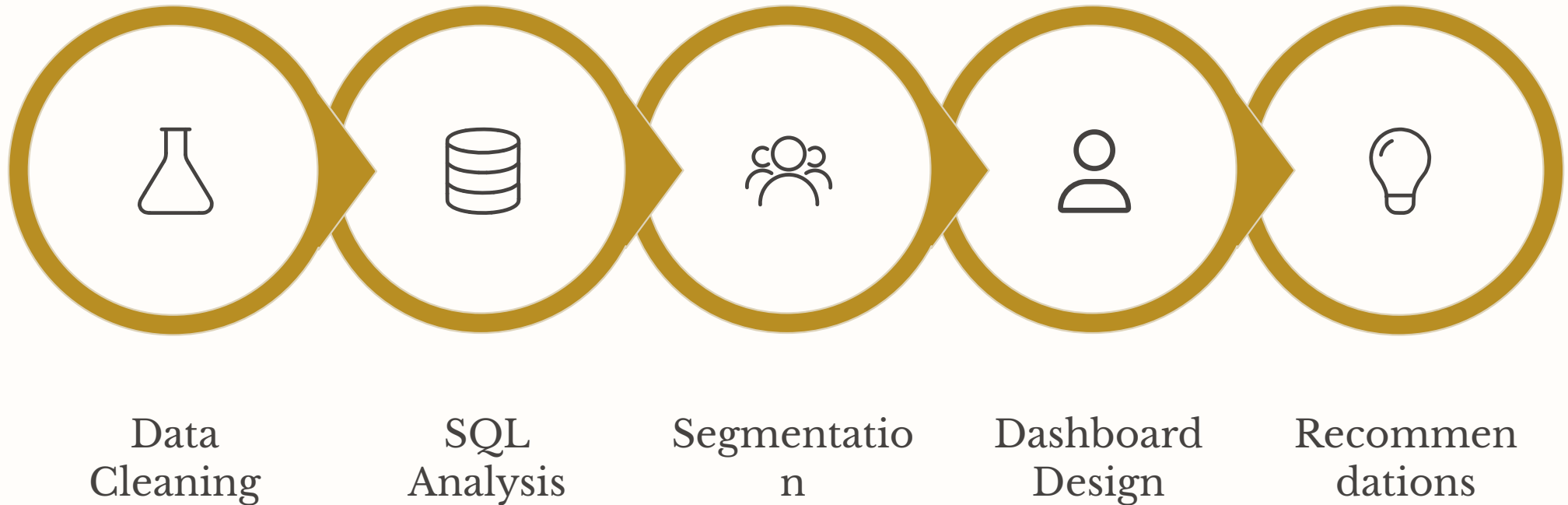
35.34%

Revenue Loss %

Proportion of total revenue wiped
out by churn

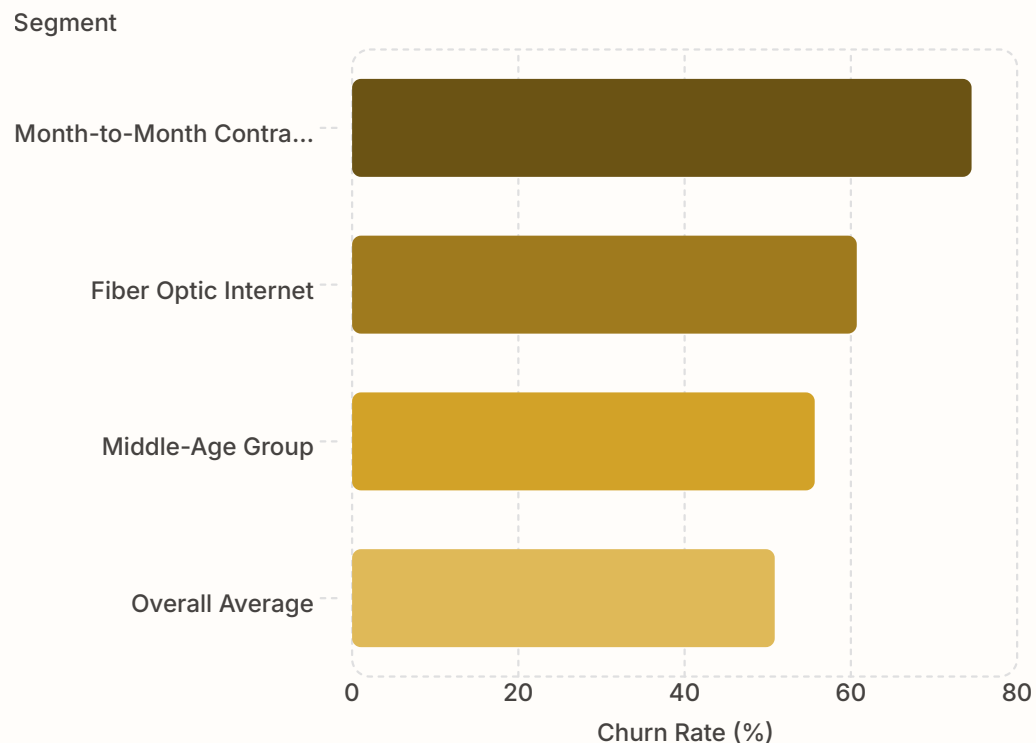
Project Methodology

A structured, end-to-end analytics workflow — from raw data to boardroom-ready insights.



Each phase builds on the last, ensuring findings are grounded in clean, validated data and communicated through clear visual storytelling.

Key Findings: Churn Behaviour



What the Data Reveals

- **1,869 customers churned** vs. 1,665 who stayed and 142 who joined
- **Month-to-Month contracts** are the highest-risk segment — nearly 3 in 4 customers leave
- **Fiber Optic** customers churn at a significantly higher rate than other internet types
- **Middle-age customers** represent the most vulnerable age cohort
- **Low satisfaction scores** are a strong early warning signal for churn risk

Revenue Impact: The Cost of Churn

Total Revenue Lost

\$3,684,459.82 in revenue attributed directly to churned customers across all segments

Revenue Loss %

35.34% of total customer revenue is at risk — over one-third of the business

Highest-Loss Category

The **Competitor** churn category is associated with the single largest revenue loss, pointing to competitive pricing pressure

Payment Method Risk

Bank Withdrawal shows the highest revenue loss percentage by payment method, suggesting billing friction



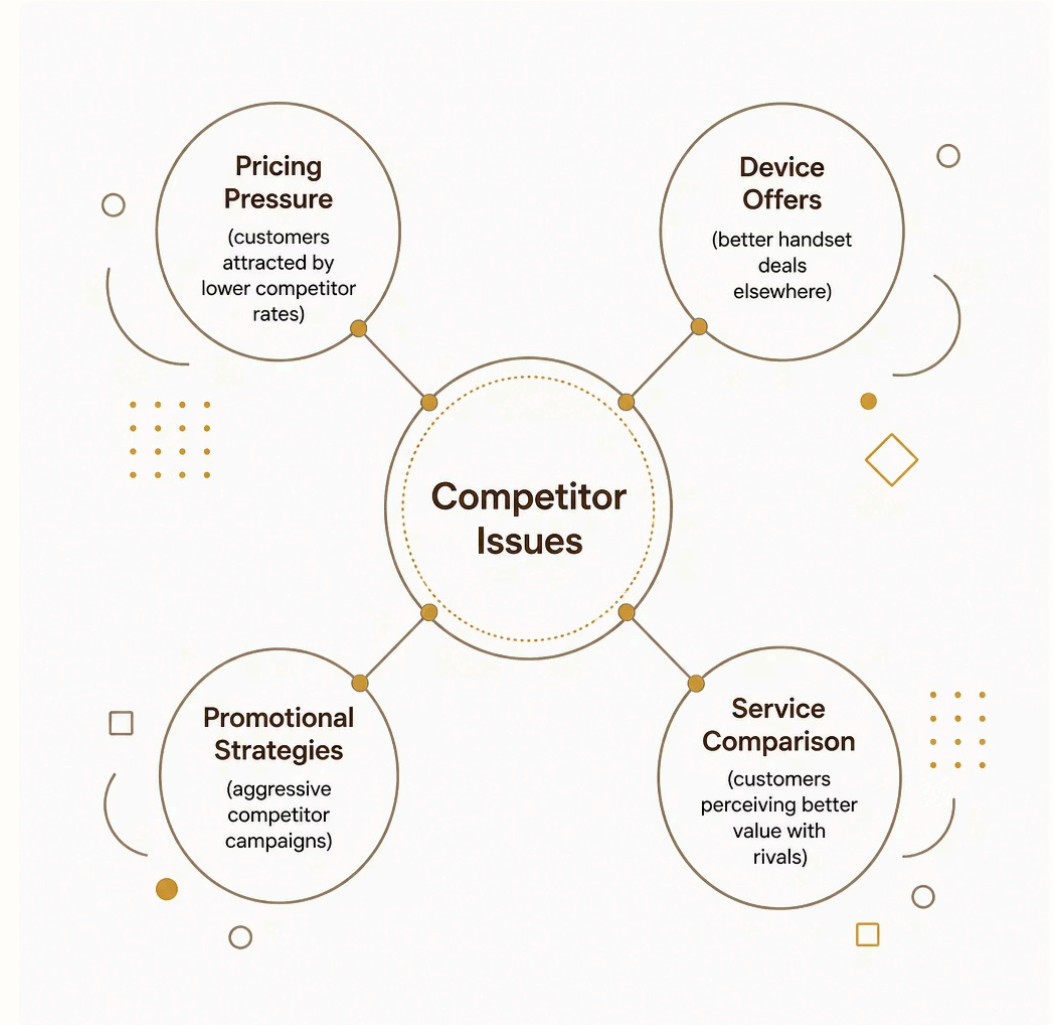
Churn Drivers: Categories & Reasons

Top Churn Categories

Competitor-related issues dominate churn categories, indicating that pricing, device offers, and promotional strategies from rivals are the primary external pressure on retention.

Customers citing competitor reasons also tend to have higher CLTV — making this category the most financially damaging.

i Competitor churn is not just the most common reason — it is the **costliest**.



Dashboard 1: Churn Analysis



What This Dashboard Covers

The Churn Analysis Dashboard provides a behavioural lens into customer attrition, enabling stakeholders to pinpoint where intervention is most urgent.

- Churn KPIs at a glance
- Churn rate by contract type, internet type, and age group
- Satisfaction score vs. churn correlation
- Top churn categories and reasons ranked by volume

Dashboard 2: Revenue Impact



What This Dashboard Covers

The Revenue Impact Dashboard translates churn into financial terms, making the business case for retention investment clear to leadership.

- Total revenue and revenue loss figures
- Revenue loss breakdown by churn category and reason
- Revenue loss percentage by payment method
- High-CLTV churned customers flagged for win-back review

Business Recommendations

1

Incentivise Contract Upgrades

Offer loyalty discounts and bundled plans to convert Month-to-Month customers to longer contracts.

2

Proactive Low-Satisfaction Outreach

Deploy service recovery and retention offers to low-satisfaction customers before they churn.

3

Improve Fiber Optic Experience

Review pricing, reliability, and support quality for Fiber Optic customers — the highest-churn internet segment.

4

Counter Competitor Pressure

Benchmark competitor pricing and promotions; develop targeted retention offers for high-CLTV at-risk customers.

5

Win-Back High-CLTV Customers

Prioritise churned high-value customers with personalised re-engagement campaigns and service improvements.

Project at a Glance

Tools & Technologies

Python

Data cleaning & preprocessing

SQL

KPIs, churn & revenue analysis

Tableau

Dashboard design & visual storytelling

GitHub

Portfolio documentation

Skills Demonstrated

- Data cleaning and exploratory analysis
- SQL querying and KPI calculation
- Customer churn and revenue-loss analysis
- Customer segmentation
- Dashboard design in Tableau
- Business insight generation and data storytelling

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Full repository available on GitHub: **telco-customer-churn-analysis**